Xin Yu

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Education

The Pennsylvania State University — Aug. 2023 – May 2027 (Expected)
PhD Candidate in Statistics, GPA: 4.0/4.0 — *State College, PA*

Nankai University — Aug. 2019 – May 2023
Bachelor of Science in Statistics, GPA: 3.90/4.0 — *Tianjin, China*

Technical Skills

- **Modeling:** Agentic AI, post-training for foundation models, reasoning models, generative recommendation, multimodal learning, large language models, diffusion models, retrieval-augmented generation
- **Systems:** Fully Sharded Data Parallel (FSDP), large-scale training, reinforcement learning for foundation models
- **Programming:** Python (PyTorch), R, MATLAB, C++, Java, SQL

Experience

Google — Aug. 18, 2026 – Nov. 17, 2026
Student Researcher — *Sunnyvale, CA*

- Worked on self-evolving agents, focusing on improving autonomous reasoning and adaptation through iterative post-training.

Meta, Feed Discovery — May 2026 – Aug. 2026
Research Scientist Intern — *Menlo Park, CA*

- Built a tool-use-augmented reasoning framework for ML development by curating a tool library and enhancing small models' agentic reasoning and execution capabilities through post-training.
- Deployed a new generative recommendation generator for the Facebook CCS product line, training it with CPT and SFT and further improving it with GRPO using latent-space reward signals derived from user interactions.

TikTok Ads Integrity — Nov. 2025 – May 2026
Research Scientist Intern — *San Jose, CA*

- Built an MLLM-based pipeline to extract structured SKU attributes from product landing-page screenshots, enabling scalable product understanding for ads integrity and downstream recommendation.
- Improved hard cases through iterative training on failure examples, including self-rewarding and on-policy RL distillation, and shipped updated checkpoints to production.

Bytedance Seed — May 2025 – Dec. 2025
Research Scientist Intern — *Seattle, WA*

- Optimized the volcengine/verl reinforcement learning framework with GRPO in FSDP, enabling lightweight and high-performance training across large-scale models.
- Developed annotation methods for ranking structure in open-source pairwise preference datasets and optimized diffusion models with list-wise human preferences.

Microsoft Research Asia — Jan. 2023 – May 2023
Intelligent Multimedia Group Intern — *Beijing, China*

- Developed lightweight convolutional architectures for visual recognition under data imbalance, improving model efficiency and generalization.
- Shipped the model to an internal OCR pipeline, enabling practical deployment for real-world OCR tasks.

Publications

ExpertWeaver: Unlocking the Inherent MoE in Dense LLMs with GLU Activation Patterns
*Ziyu Zhao, Tong Zhu, **Xin Yu**, Zhi Zhang, Tiantian Fan, Jinluan Yang, Kun Kuang, Zhongyu Wei, Fei Wu, Yu Cheng*
Forty-third International Conference on Machine Learning (ICML 2026)

Towards Better Optimization for Listwise Preference in Diffusion Models
Jiamu Bai, **Xin Yu***, *Meilong Xu, Weitao Lu, Xin Pan, Kiwan Maeng, Daniel Kifer, Jian Wang, Yu Wang*
International Conference on Learning Representations (ICLR 2026)

Each Rank Could Be an Expert: Single-Ranked Mixture of Experts LoRA for Multi-task Learning
*Ziyu Zhao, Yixiao Zhou, **Xin Yu**, Zhi Zhang, Didi Zhu, Tao Shen, Zexi Li, Jinluan Yang, Xuwu Wang, Jing Su, Kun Kuang, Zhongyu Wei, Fei Wu, Yu Cheng*
SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2026)

AltLoRA: Towards Better Gradient Approximation in Low-Rank Adaptation with Alternating Projections
***Xin Yu**, Yujia Wang, Jinghui Chen, Lingzhou Xue*
Neural Information Processing Systems (NeurIPS 2025)

Understanding the Statistical Accuracy-Communication Trade-off in Personalized Federated Learning
***Xin Yu**, Zelin He, Ying Sun, Lingzhou Xue, Runze Li*
International Conference on Machine Learning (ICML 2025)

Preprint

A New Inexact Manifold Proximal Linear Algorithm with Adaptive Stopping Criteria
*Zhong Zheng, **Xin Yu**, Shiqian Ma, Lingzhou Xue*
Minor revision informs optimization

PrunedLoRA: Efficient and Robust Gradient-Based Structural Pruning for Low-Rank Adaptation
***Xin Yu**, Cong Xie, Ziyu Zhao, Tiantian Fan, Lingzhou Xue, Zhi Zhang*
Submitted to NeurIPS 2026

EXACT: Explicit Attribute-Guided Decoding-Time Personalization
***Xin Yu**, Hanwen Xing, Lingzhou Xue*
Submitted to NeurIPS 2026

Preference-Based Self-Distillation: Beyond KL Matching via Reward Regularization
***Xin Yu**, Liucheng Liao, Yiwen Zhang, Yingchen Yu, Lingzhou Xue, Qinzhen Guo*
Submitted to NeurIPS 2026

Awards

- Machine Learning Summer School Travel Award, Columbia University, Jun. 19–27, 2026
- Best Paper, C.R. Conference, 2025
- Jack and Eleanor Pettit Scholarship in Science, 2024–2025
- University Graduate Fellowship, 2023–2024
- Second Prize, National Undergraduate Academic Competition, 2023
- Second Prize, National High School Mathematics Competition, 2018
- Second Prize, National High School Physics Competition, 2018